SELECTION 2

Team Teaching Basic Programming

2026

Learning Outcome

- Students are able to understand the definition and purpose of the nested selection syntax
- Students understand the basic structure of the nested selection syntax
- Students are able to solve problems by creating a Java program that uses the nested selection syntax

Nested Selection

- Nested selection (NESTED IF) is a type of selection used to make decisions in levels (multiple layers)
- Inside an IF (or IF-ELSE) statement, there can be another IF (or IF-ELSE) statement

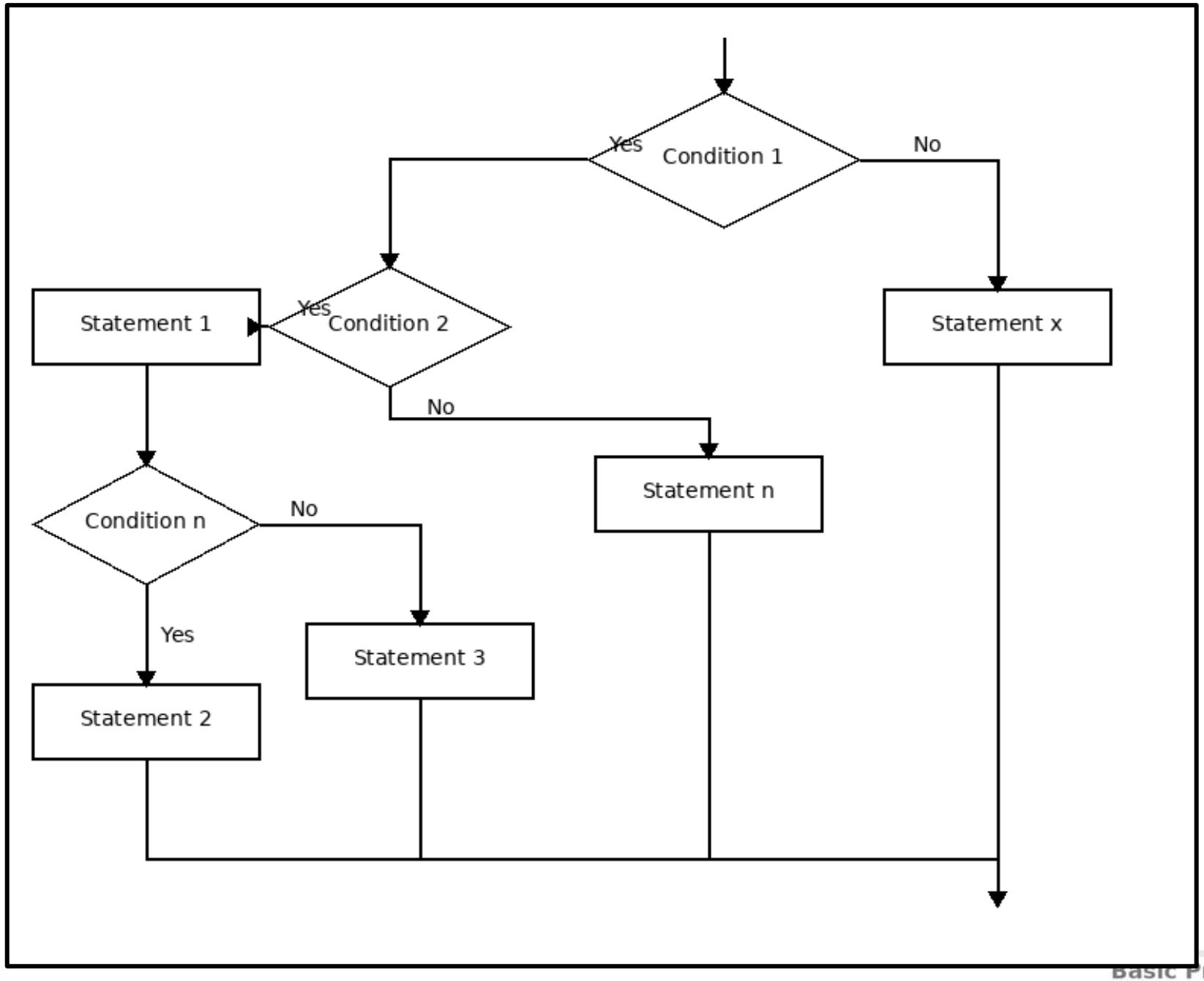
General Form of Nested Selection:

```
if (condition 1){
        if (condition 2){
                statement 1;
                …
                …
                if (condition n){
                                statement 2;
                } else {
                        statement 3;
                }
        } else {
                statement n;
        }
} else {
        statement x;
}
```

Nested Selection

- The condition evaluated first is the outermost IF condition (condition 1).
- If condition 1 is false, the outer ELSE statement (paired with that IF) is processed. However, if that ELSE statement is not written, the condition check stops.
- If condition 1 turns out to be true, the next, deeper condition (condition 2) is evaluated. If condition 2 is false, the ELSE statement (paired with that IF) is processed.
- However, if that ELSE statement is not written, the condition check stops.

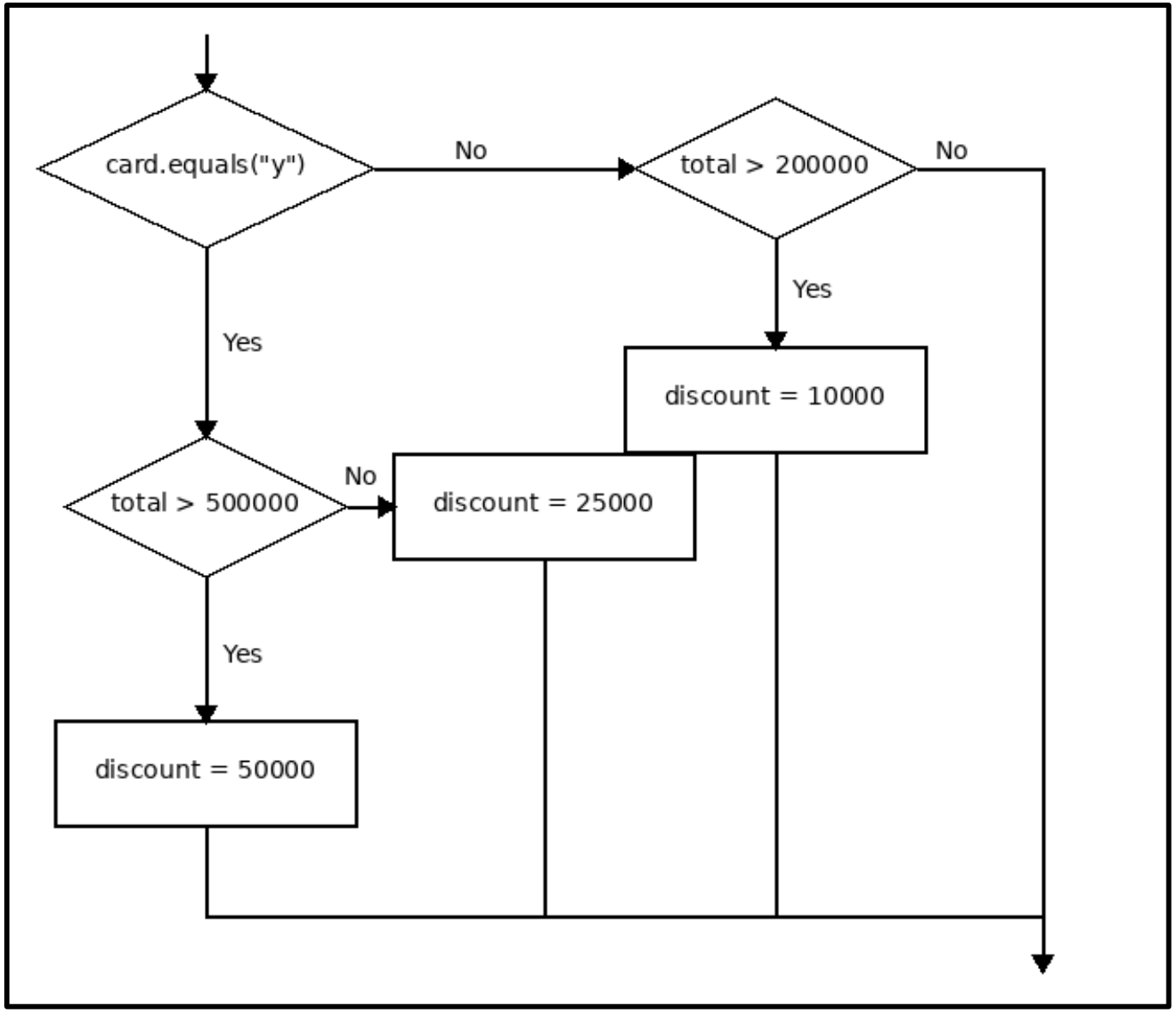
Flowchart of Nested Selection:



Example of Using Nested Selection

- When a customer pays at the cashier, the cashier asks:
- **Does the customer have a membership card?**
- TRUE:
  - **Is the total price of the items more than Rp 500,000?**
    - TRUE: The customer gets a discount of Rp 50,000
    - FALSE: The customer gets a discount of Rp 25,000
- FALSE:
  - **Is the total price of the items more than Rp 200,000?**
    - TRUE: The customer gets a discount of Rp 10,000
    - FALSE: The customer does not get a discount

Example Flowchart:



Example Program:

```java
import java.util.Scanner;
public class cashier {
    public static void main(String[] args) {
        int total, discount, payment;
        String card;
        Scanner sc = new Scanner (System.in);
        System.out.print("Does the customer have a membership card (y or n)? ");
        card = sc.nextLine();
        System.out.print("What is the total price of the items? Rp ");
        total = sc.nextInt();
        if (card.equals("y")) {
            if (total > 500000) {
                discount = 50000;
            } else {
                discount = 25000;
            }
        } else {
            if (total > 200000) {
                discount = 10000;
            } else {
                discount = 0;
            }
        }
        payment = total - discount;
        System.out.println("Total amount to be paid: Rp " + payment);
    }
}
```

Logical Expression

- There are 3 types of logical operators used in IF-ELSE statements:

  && : AND

  || : OR

  ! : NOT

- **Logical expression** is an expression that uses one or more logical operators.
- The operators applied in a logical expression are evaluated from **left to right**

Logical Expression

- When evaluating **(e_1 && e_2)**, if e_1 results in FALSE, then e_2 is not evaluated. Therefore, the entire value of the expression (e_1 && e_2) is considered false
- However, if e_1 results in TRUE, then e_2 is then evaluated to determine the value of the whole expression
- Example:
- `If(speed == 0 && engineOn == true)`
  - `System.out.println("Turn off the engine");`

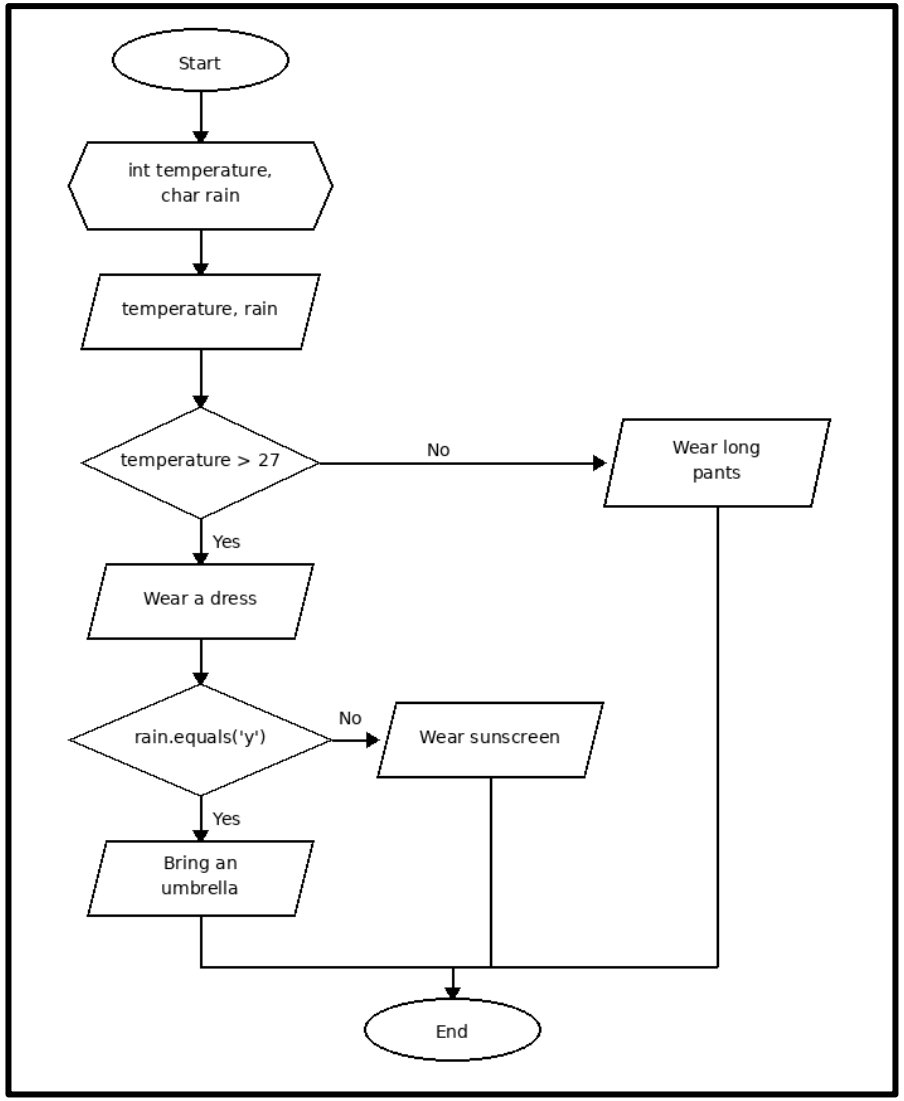
Logical Expression

- When evaluating **(e_1 || e_2)**, if e_1 results in TRUE, then e_2 is not evaluated. Therefore, the entire value of the expression (e_1 || e_2) is considered true
- However, if e_1 results in FALSE, then e_2 is then evaluated to determine the value of the whole expression
- Example:
- `If(speed == 0 || engineOn == true)`
  - `System.out.println("Turn off the engine");`

Example Case Study

- A system is built to determine the clothing and equipment a user should bring based on the weather. If the temperature is more than 27°C, the user is advised to wear a dress, then a check is made on whether it is currently raining. If it is raining, the user is advised to bring an umbrella; if not, the user is advised to wear sunscreen. However, if the temperature is less than or equal to 27°C, the user is advised to wear long pants
- Create a flowchart for that system!

Case Study Flowchart:



EXERCISE

Exercise 1

Arrange an algorithm using a flowchart to determine the largest number from 3 input numbers (without using logical operators)

input:

num1 = 28
num2 = 54
num3 = 15

output:

largest number : 54

Exercise 2

- Every Wednesday, a bookstore gives discounts to its customers depending on the type of book purchased
- A 10% discount is given if the book purchased is a dictionary; an additional 2% discount is given if more than 2 books are purchased
- A 7% discount is given if the book purchased is a novel; an additional 2% discount is given if more than 3 novels are purchased, while if 3 or fewer novels are purchased, an additional 1% discount is given
- Customers get a 5% discount on books other than dictionaries and novels if more than 3 books are purchased
- Create a flowchart (use logical operators) to determine the total amount to be paid if the input is the type and number of books, and the output is the discount amount

Exercise 3

A shoe store has a price list based on brand, category, and size. The data is shown in the table below

Shoe Code	Brand	Category	Size	Price
1	Converse	Slip On	36-40	800.000
		High Top	40-44	1.200.000
2	Sketcher	Woman	36-41	1.000.000
		Man	41-44	1.800.000
3	Nike	Kids	36-40	750.000
		Adult	40-44	1.500.000

Create a flowchart (without using logical operators) to determine the price to be paid if the input is the brand, category, and size. The output is the price of the shoes.